

Shivam Raj

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Summary

Ph.D. candidate in experimental high-energy physics on the CMS experiment at CERN. Research interests span BSM resonance searches, statistical methods for limit-setting, and machine learning for detector reconstruction and data quality monitoring. Seeking a postdoctoral position to continue this work within a CMS (or comparable LHC) research group.

- Leading CMS Run 3 resonant search $X \rightarrow YH \rightarrow b\bar{b}\gamma\gamma$ (B2G-25-010) end to end — signal sample production and validation, categorization, statistical framework, and multi-committee review — heading toward pre-approval.
- B2G contact and formal reviewer for the CMS Muon Physics Object Group (POG).
- Core contributor to two CMS-wide papers: quantization & ONNX deployment for the MLPF algorithm, and institutional review of an anomaly-detection methodology.
- Participated directly in CMS Run 3 data-taking as Pixel Detector-on-Call, monitoring detector status and responding to operational issues in real time.

Education

Ph.D. in Physics

2022 – Expected 2027

Department of Physics, The Catholic University of America

GPA: 3.93/4.0

Advisors: Dr. Rachel Bartek, Dr. Shin-Shan Yu & Prof. Aaron Dominguez. Focus: BSM resonance searches, statistical methods for limit-setting, machine learning for detector reconstruction and data quality monitoring.

Integrated M.Sc., Physics

2017–2022

National Institute of Science Education and Research (NISER)

Bhubaneswar, Odisha, India. Advisor: Dr. Prolay Kr. Mal.

Technical Skills

- **Programming:** Python, C++, C, Bash, Git
- **ML / DL:** PyTorch, TensorFlow, Keras, ONNX (quantization & deployment), Graph Neural Networks
- **Statistics & UQ:** Conformal prediction (MAPIE, crepes), iterative Bayesian unfolding, bootstrap resampling
- **HEP Analysis Software:** ROOT, PyROOT, CMSSW, uproot, awkward-array, Pythia8, GEANT4, MadGraph, Combine
- **CMS Computing & Grid:** CERN lxplus/AFS/EOS, HTCondor, CVMFS, DAS (dasgoclient), brilcalc
- **Data Analysis:** NumPy, Pandas, Origin
- **Other:** $\text{\LaTeX}$, Mathematica, Linux

Research Experience

Lead Analyzer (05/2024–Present), CMS Collaboration — Search for $X \rightarrow YH \rightarrow b\bar{b}\gamma\gamma$ (B2G-25-010), CERN, Switzerland

- Leading a Run 3 search for resonant scalar pair production ($Y \rightarrow b\bar{b}, H \rightarrow \gamma\gamma$) in the NMSSM, using 172 fb⁻¹ of 2022–2024 data; successor to the Run 2 analysis.
- Built the end-to-end analysis chain — signal categorization, parametric signal and background modeling, and CL_s limit extraction — across a ~200-point mass grid per data-taking era, with automated multi-year combination and batch orchestration over HTCondor.
- Validated the statistical fit framework ahead of pre-approval by resolving a ~21σ false-signal bias to within the standard acceptance threshold and fixing two independent workspace bugs (cross-year parameter collisions; an unphysical fit-range default in the highest-purity categories, S/B up to ~29,000:1).
- Diagnosed a mass-degeneracy between the $Y \rightarrow b\bar{b}$ resonance and the SM Higgs boson at $m_Y = 125$ GeV, confirmed the full three-year combination remains clean at this point, and proposed an analyzer-level fix for the next iteration of the analysis.
- Guiding the analysis through concurrent CMS review (Generator, Statistics, PPD, EGamma, JetMET/JERC); head-

ing toward pre-approval, with a paper to follow.

B2G Contact & Formal Reviewer (2025–Present), CMS Muon Physics Object Group (POG), CERN, Switzerland

- Serve as the B2G analysis group’s liaison to the Muon POG and conduct formal object-level reviews of muon reconstruction, identification, and scale-factor treatment for analyses submitted to the collaboration.
- Completed muon-sector reviews for more than 20 CMS analyses, most recently B2G-26-005 (resonant $X \rightarrow HY \rightarrow b\bar{b}WW$, single-lepton search): audited object definitions, corrections, and systematics against the review questionnaire and authored formal reviewer comments.

Pixel Detector-on-Call (06/2025–12/2025), CMS Collaboration — Run 3 Detector Operations, CERN, Switzerland

- Participated directly in CMS Run 3 data-taking as the on-call operations expert for the Pixel sub-detector: monitored detector and data-quality status in real time and responded to operational issues during collisions.

Graduate Researcher (05/2023–Present), Pixel Detector Development for the CMS Inner Tracker, Catholic University of America, Washington, DC

Supervisors: *Dr. Rachel Bartek, Prof. Aaron Dominguez*

- Assembled 1×2 and 2×2 TFPX (Forward Pixel) modules with CROC v2 readout chips and $25 \times 100 \mu\text{m}^2$ -pixel silicon sensors for the CMS Phase-2 Inner Tracker upgrade, as part of one of four US CMS production sites (with Purdue, Nebraska, and Boston University).
- Performed precision module assembly: gluing High-Density Interconnects (HDI) to readout chips on an Aerotech A3200 gantry system, and wire-bonding HDI-to-ROC interconnections ($25 \mu\text{m}$ Al wire, Hesse BJ855 bonder); validated bond quality via pull testing (typical strength ~ 9 cN).
- Conducted module electrical qualification — IV curves, PixelAlive, and Shunt-LDO tests — in a temperature-controlled ($\sim -10^\circ\text{C}$) cold box to characterize sensor leakage current and confirm module health ahead of shipment to partner testing sites (UIC, Ohio State).

Graduate Researcher (08/2024–12/2024), ML-Based Data Quality Monitoring, CMS EGamma, CERN, Switzerland

- Developed an autoencoder-based anomaly-detection pipeline for offline certification of EGamma data, with an online (real-time) deployment during Run 3 data-taking to flag anomalous events; served as CMS institutional reviewer for the resulting methodology paper.

Graduate Researcher (07/2023–07/2025), Post-Training Quantization of the ML Particle-Flow (MLPF) Algorithm *With Dr. Joosep Pata, Dr. Javier Duarte, Farouk Mokhtar, David Southwick, Eric Wulf*

- Developed a post-training quantization and ONNX deployment pipeline to reduce the size and inference latency of the graph-neural-network-based MLPF reconstruction algorithm.
- Contribution incorporated into the CMS-wide “Full Event Interpretation with ML-Based Particle-Flow Reconstruction” paper (*Eur. Phys. J. C*, in press).

Graduate Researcher (05/2023–08/2024), Tracker Material Validation & Inner Tracker R&D, Catholic University of America, Washington, DC

Supervisors: *Dr. Rachel Bartek, Prof. Aaron Dominguez*

- Validated detector material budgets in simulation across CMS Phase 0–2 tracker upgrade geometries; contributed to CMSSW material-validation software.
- Hands-on pixel-detector hardware R&D for the HL-LHC inner tracker: sensor gluing, wire-bonding, pull testing, and pixel-alive qualification.

Master’s Thesis & Undergraduate Research (2020–2022), NISER, Bhubaneswar, India

Primary supervisor: *Dr. Prolay Kr. Mal*

- Thesis: DNN-based search for heavy vector-like quark / T' resonances against SM Higgs and non-resonant backgrounds; benchmarked against a mature BDT-based CMS analysis using the Higgs Combined Tool.
- Earlier projects: Kalman-filter track fitting for an RPC cosmic-ray muon detector (Prof. Bedangadas Mohanty); two- and three-flavor neutrino oscillation phenomenology (Prof. Sanjay Kr. Swain); $e^+e^- \rightarrow \mu^+\mu^-$ kinematics with Pythia/ROOT.

Selected Publications

Full list: [INSPIRE-HEP](#) | [Google Scholar](#)

1. CMS Collaboration, "Full Event Interpretation with ML-Based Particle-Flow Reconstruction in the CMS Detector," *Eur. Phys. J. C* (2026), DOI:10.1140/epjc/s10052-026-15754-5, [arXiv:2601.17554](#). *Contribution: model development; designed and implemented the post-training quantization framework and ONNX deployment pipeline.*
2. CMS Collaboration, "Wasserstein-Normalized Autoencoder for Anomaly Detection," *Mach. Learn.: Sci. Technol.* (2026), [arXiv:2510.02168](#). *Role: CMS institutional reviewer.*
3. M. Agarwal et al., "Applications of Deep Learning to Physics Workflows," 2023, [arXiv:2306.08106](#).

Selected Talks and Posters

- [Poster](#): USCMS Annual Collaboration Meeting 2026, University of Maryland, College Park, MD — June 2026.
- [Talk](#): USCMS Annual Collaboration Meeting 2025, Rice University, Houston, TX — May 2025.
- [Poster](#): USCMS Annual Collaboration Meeting 2025, Rice University, Houston, TX — May 2025.
- [Talk](#): TFPX 2025 Workshop, Rice University, Houston, TX — May 2025.
- [Poster](#): CMS Upgrade Week, CERN, Geneva, Switzerland — September 2024.
- [Poster](#): USCMS Annual Collaboration Meeting 2024, Princeton University, Princeton, NJ — June 2024.
- [Poster](#): University Research Day, Catholic University of America, Washington, DC — April 2024.

Teaching Experience

- Teaching Assistant, Dept. of Physics, The Catholic University of America — College Physics (Fall 2022), University Physics (Mechanics & Electronics, incl. Honors sections), and Introductory Physics Labs; 4 semesters (Fall 2022 – Spring 2024).

Honors & Awards

- All-India Rank 32, Graduate Aptitude Test in Engineering (GATE), Physics.
- Selected, LHC Early Career Mentorship Program (2021–2022).
- INSPIRE Scholarship (2017–2022), Dept. of Science & Technology, Government of India — merit-based fellowship for research in the basic sciences.

Summer Schools & Training

- CTEQ Summer School on QCD and Electroweak Phenomenology, University of Münster, Germany — 2024.
- SLAC Summer Institute, Palo Alto, CA, USA — 2023.
- Summer of Quantum (SoQ) Short Course, LPS, University of Maryland, College Park, MD, USA — 2023.